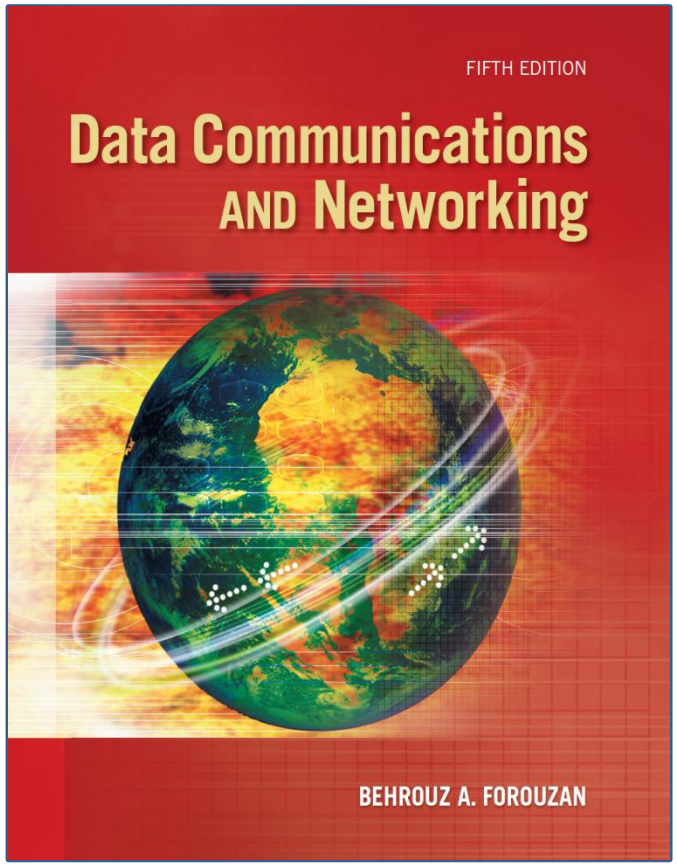


Network Services

Lecturer: Dr. Khaled BinAbdl

Lecture 3

Domain Name Caching

- When a server **asks** for a **mapping** from **another** server and **receives** the **response**, it **stores** this information in its **cache** memory before sending it to the **client**.

- To **inform** the client that the **response** is coming from the **cache** memory and not from an **authoritative** source, the server marks the response as ***unauthoritative***.

Domain Name Caching cont.

- Caching **speeds up** resolution, but it can also be **problematic**.
- If a server **caches** a mapping for a **long time**, it may send an **outdated** mapping to the client.
- To **counter** this, **two** techniques are used:

Domain Name Caching cont.

- **First**, the **authoritative** server always **adds** information to the mapping called ***time to live*** (**TTL**).
- It defines the time in **seconds** that the **receiving** server **can** cache the information.
- **After** that time, the mapping is **invalid** and any **query** must be sent **again** to the **authoritative** server.

Domain Name Caching cont.

- **Second**, DNS requires that each server keeps a **TTL** counter for each mapping it caches.

- The **cache** memory must be **searched periodically** and those mappings with an **expired** TTL must be **purged**.

DNS Resource Records

- The **name server** stores a **database** of ***resource records***.
- A *resource record* is a **5-tuple** structure:
- **(Domain Name, Type, Class, TTL, Value)**

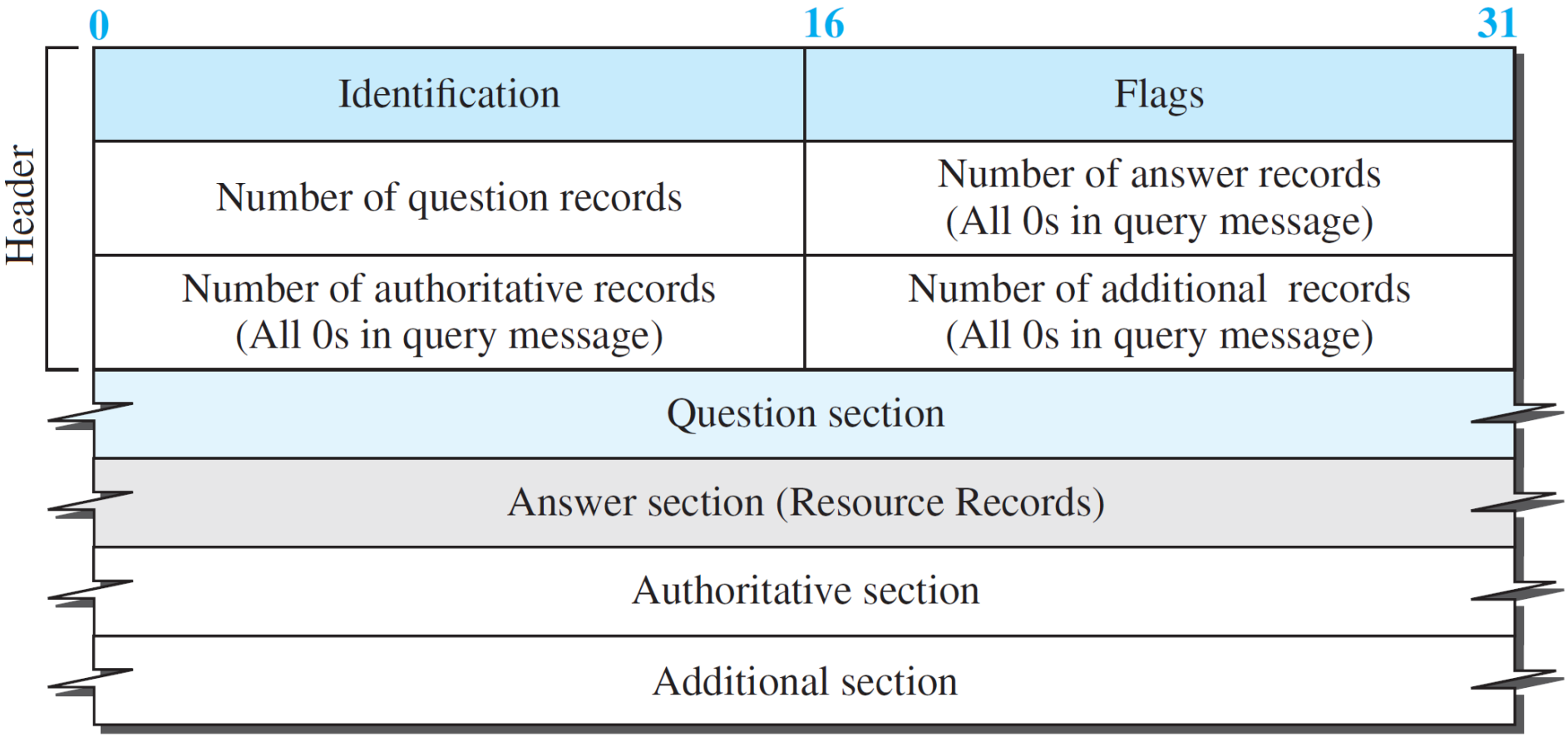
DNS Resource Records cont.

- **Domain** name: is what identifies the resource record.
- **Value**: is the information kept about the domain name.
- **TTL**: is the number of seconds for which the information is valid.
- **Class**: the type of network; we are only interested in the class **IN** (Internet).
- **Type**: is how the value should be interpreted.

DNS Resource Records cont.

Type	*Interpretation of value*
A	A 32-bit IPv4 address (see Chapter 18)
NS	Identifies the authoritative servers for a zone
CNAME	Defines an alias for the official name of a host
SOA	Marks the beginning of a zone
MX	Redirects mail to a mail server
AAAA	An IPv6 address (see Chapter 22)

DNS Messages



Dynamic Domain Name System (DDNS)

- In **DDNS**, when a **binding** between a name and an address is determined, the information is **sent**, usually by DHCP to a **primary** DNS server.
- The **primary** server updates the **zone**.
- The **secondary** servers are **notified** either **actively** or **passively**.
  - In **active** notification, the **primary** server **sends** a message to the **secondary** servers about the **change** in the zone.
  - In **passive** notification, the **secondary** servers **periodically check** for any **changes**.

DNS Security

DNS can be attacked in several ways including:

- The **attacker** may **read** the **response** of a DNS server to find the **nature** or names of sites the user **mostly** accesses.
- The attacker may **intercept** the **response** of a DNS server and **change** it or create a totally new **bogus (fake)** response to direct the user to the **site** or domain the attacker wishes the user to access.
- The attacker may **flood** the DNS **server** to **overwhelm** it or eventually **crash** it.

Dynamic Host Configuration Protocol (DHCP)

– Address **assignment** in an organization can be done **automatically** using the **DHCP.**

– **DHCP** is an application-layer **program**, using the **client-server** paradigm, that actually helps TCP/IP at the network layer.

– **DHCP** often called a *plugand-play protocol*.

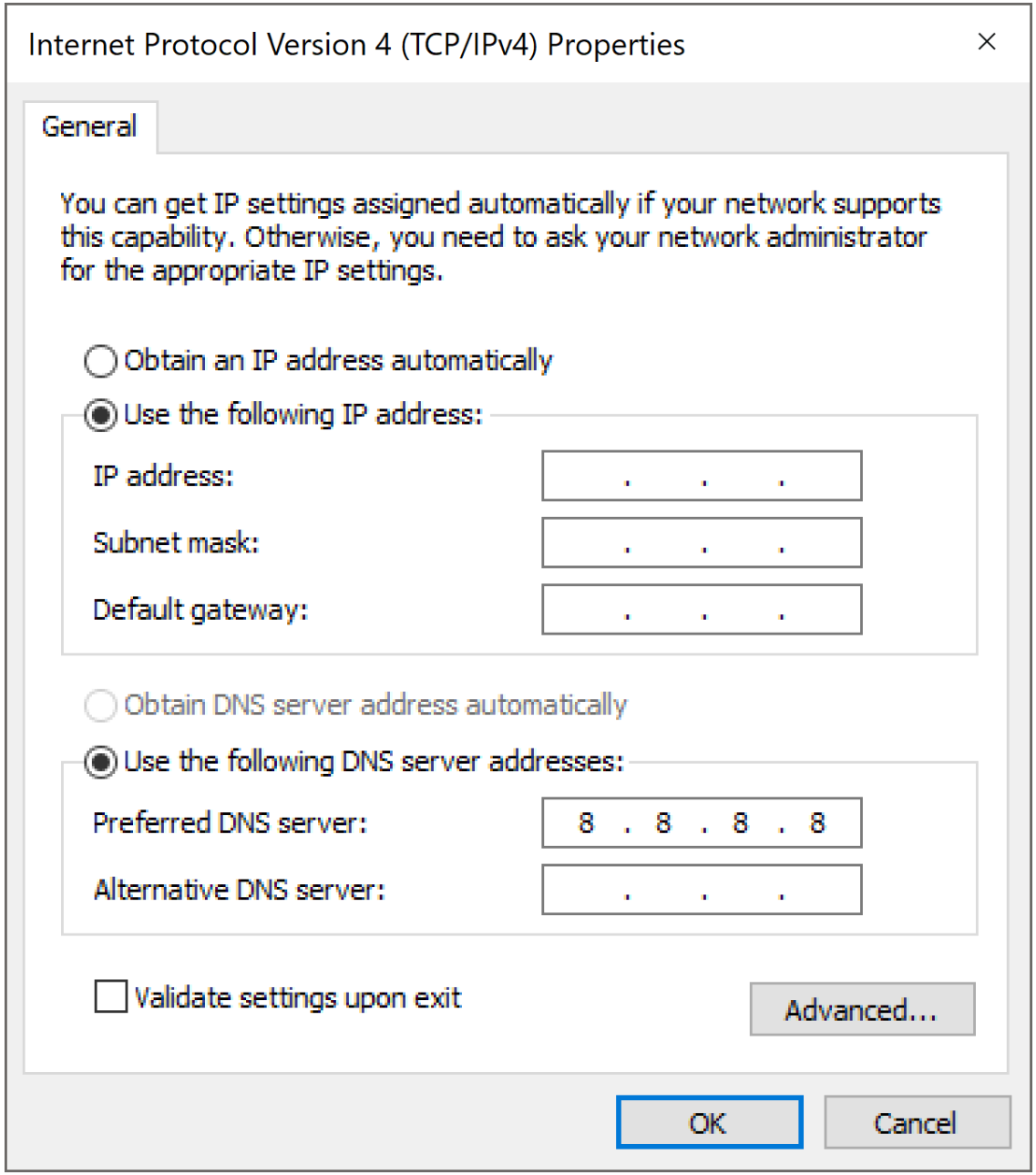
DHCP Usage

- A network **manager** can configure **DHCP** to assign **permanent** IP addresses to the **host** and **routers**.
- **DHCP** can also be configured to provide **temporary**, on demand, IP addresses to hosts.
  - Can provide a **temporary** IP address to a **traveller**.
  - ISP with **1000** granted addresses to provide services to **4000** households, assuming not more than **1/4** of **customers** use the Internet **at the same time**.

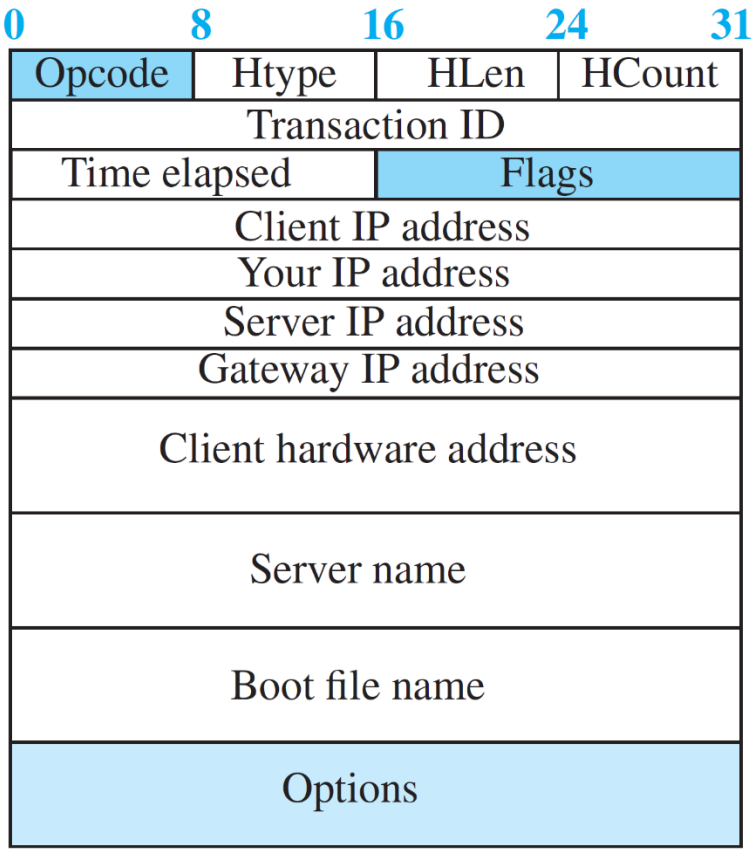
DHCP Usage cont.

– **Four** pieces of **information** are normally needed by any **host** and can be **provided** by DHCP:

1. Computer address (**IP**).
2. Prefix (**Mask**).
3. Address of a router (**Gateway**).
4. IP address of a name server (**DNS**).


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 8 . 8 . 8 . 8
Alternative DNS server:
Validate settings upon exit
Advanced...
OK
Cancel

DHCP Message Format



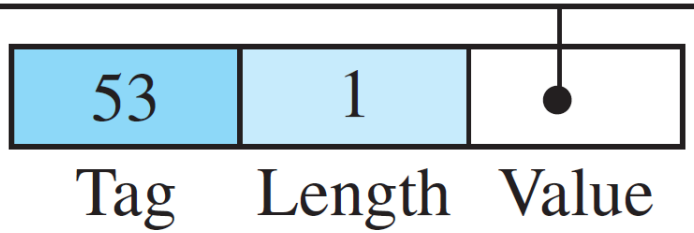
Fields:

Opcode: Operation code, request (1) or reply (2)
Htype: Hardware type (Ethernet, ...)
HLen: Length of hardware address
HCount: Maximum number of hops the packet can travel
Transaction ID: An integer set by the client and repeated by the server
Time elapsed: The number of seconds since the client started to boot
Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used
Client IP address: Set to 0 if the client does not know it
Your IP address: The client IP address sent by the server
Server IP address: A broadcast IP address if client does not know it
Gateway IP address: The address of default router
Server name: A 64-byte domain name of the server
Boot file name: A 128-byte file name holding extra information
Options: A 64-byte field with dual purpose described in text

DHCP Message Options Field

Option format

1 DHCPDISCOVER	5 DHCPACK
2 DHCPOFFER	6 DHCPNACK
3 DHCPREQUEST	7 DHCPRELEASE
4 DHCPDECLINE	8 DHCPINFORM



DHCP Operation

Client
IP Address: ?

Server
IP Address: 181.14.16.170

DHCPDISCOVER

Transaction ID: 1001
Lease time:
Client address:
Your address:
Server address:

Source port: 68 Destination port: 67

Source address: 0.0.0.0
Destination address: 255.255.255.255.

DHCPOFFER

Transaction ID: 1001
Lease time: 3600
Client address:
Your address: 181.14.16.182
Server address: 181.14.16.170

Source port: 67 Destination port: 68

Source address: 181.14.16.170
Destination address: 255.255.255.255.

DHCPREQUEST

Transaction ID: 1001
Lease time: 3600
Client address: 181.14.16.182
Your address:
Server address: 181.14.16.170

Source port: 68 Destination port: 67

Source address: 181.14.16.182
Destination address: 255.255.255.255.

DHCPACK

Transaction ID: 1001
Lease time: 3600
Client address:
Your address: 181.14.16.182
Server address: 181.14.16.170

Source port: 67 Destination port: 68

Source address: 181.14.16.170
Destination address: 255.255.255.255.

Legend

Application
UDP
IP

Note:
Only partial information is given.

Automatic Private IP Addressing (APIPA)

- To get an IPv4 address, the **client** will **select** an address at **random** in the range **169.254.1.0** to **169.254.254.255**, with a netmask of **255.255.0.0**.
- The client will then **send** an **ARP** packet **asking** for the **MAC** address that corresponds to the randomly-generated **IPv4** address.
- If any other **machine** is **using** that **address**, the client will generate **another random** address and **try again**.
- **APIPA** is **implemented** in Windows **98** and **later**.